# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 2: **Acids, Bases and Salts** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** Assertion (A): Dry HCl gas does not change the colour of dry blue litmus paper. Reason (R): HCl gas does not ionise in the absence of water to release $H^+$ ions. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**2.** Equal volumes of 0.1 M HCl and 0.1 M NaOH are mixed. The resulting solution has pH: 1

**(a)** 1 **(b)** 3 **(c)** 7 **(d)** 13

**3.** A white powder X on adding water forms a hard solid mass. X is: 1

**(a)** baking soda **(b)** plaster of Paris **(c)** bleaching powder **(d)** washing soda

**4.** Assertion (A): Acid should always be added to water and not water to acid. Reason (R): Dilution of acid is a highly exothermic process. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is false but R is true.
**(d)** A is true but R is false.

**5.** 20 mL of NaOH is neutralised by 10 mL of HCl. To neutralise 50 mL of the same NaOH, the volume of HCl needed is: 1

**(a)** 5 mL **(b)** 10 mL **(c)** 100 mL **(d)** 25 mL

**6.** A salt X, when heated, loses water and turns from blue to white. X is: 1

**(a)** $Na_2CO_3 \cdot 10\,H_2O$ **(b)** $CaSO_4 \cdot 2\,H_2O$ **(c)** $CuSO_4 \cdot 5\,H_2O$ **(d)** $FeSO_4 \cdot 7\,H_2O$

**7.** Assertion (A): Aqueous solutions of glucose and alcohol do not conduct electricity. Reason (R): They do not ionise in water to give $H^+$ ions. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is false but R is true.
**(d)** A is true but R is false.

**8.** The pH of a solution is 4. The pH after diluting 10 times is approximately: 1

**(a)** 3 **(b)** 5 **(c)** 4 **(d)** 6

**9.** A compound on heating gives $CO_2$; its aqueous solution is mildly basic and is used in baking. It is: 1

**(a)** NaOH **(b)** $NaHCO_3$ **(c)** NaCl **(d)** $Na_2CO_3$

**10.** Assertion (A): Sodium hydrogen carbonate is used as an antacid. Reason (R): It reacts with HCl in the stomach to release $CO_2$ and neutralise excess acid. 1

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is false but R is true.

**(d)** A is true but R is false.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** When a few drops of conc. HCl are put on a piece of limestone, bubbles form. Name the gas, test and equation. 2

**12.** How much gypsum (172 g) heated gives how much plaster of Paris? Give the mass of water lost. 2

**13.** Why is $H_2SO_4$ never diluted in a glass bottle by pouring water on it? 2

**14.** A solution of pH 3 is diluted 100 times. Predict its pH and state its acidity. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A student is given solutions of pH 1, 4, 7, 10 and 13. Which will turn phenolphthalein pink and which can neutralise a base? Arrange by acidity. 3

**16.** A white solid X on heating gives $CO_2$ and a residue Y. Y dissolves in water, is basic and is used in whitewash. Identify X and Y and write equations. 3

**17.** Explain why curd should not be kept in copper or brass vessels but tomato ketchup can be stored in glass. 3

**OR**

A farmer's soil has pH 5. Name one chemical he can add, why, and the effect on crop. 3

**18.** Explain the use of a pH meter or universal indicator to differentiate between a strong and weak acid of the same concentration. 3

**OR**

Why does a nettle sting cause pain? Suggest and explain two remedies. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Compare the properties of HCl and $CH_3COOH$ of the same concentration with respect to pH, conductivity and reaction with Mg. Explain. 5

**OR**

Write a note on water of crystallisation, giving three examples with colour change on heating and one test. 5

**20.** A student mixes 25 mL of 0.1 M NaOH with 25 mL of 0.1 M HCl and 25 mL of 0.1 M NaOH with 50 mL of 0.1 M HCl. Predict pH regions and explain. 5

**OR**

Explain the importance of keeping the pH of the human body in the range of 7.0 to 7.8 and how the stomach and mouth contrast. 5

## END OF QUESTION PAPER — Total: 40 Marks